Non-Normal returns and estimation error in mean–variance portfolios

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Abstract

Daily stock returns have heavy tails, clustered volatility and correlations that rise in crises, yet mean–variance optimisation assumes that means and covariances are known. Using 25 large US stocks and the S&P 500 from 2000 to 2025, I document these departures from Normality, derive the Markowitz solution by Lagrange multipliers and verify it numerically, measure estimation error with random matrix theory and a simulation, and compare 1/N, minimum-variance and shrinkage portfolios in a walk-forward backtest with transaction costs. No optimised strategy reliably beat 1/N out of sample. Ledoit–Wolf shrinkage and a long-only constraint both improved on the sample covariance, but the differences are within the sampling noise of the Sharpe ratios.

1 Question and data

The question is how non-Normal real returns are, and whether mean–variance optimisation survives estimation error. The data are daily adjusted closing prices (adjusted for splits and dividends) downloaded from Yahoo Finance with `yfinance` on 11 October 2026 for 25 US large-cap stocks and the S&P 500 index: 6,538 prices from 3 January 2000 to 30 December 2025, hence 6,537 daily returns. Log returns are used for statistics and simple returns for portfolio arithmetic. Means are annualised by 252 and volatilities by $\sqrt{252}$.

The data were checked before use. All 26 series are complete from the first date, no series has a run of five or more identical prices, and there is one calendar gap longer than five days (17 September 2001, after the post-attack market closure). About 64 daily moves exceed 20% in absolute log return. They cluster at known events, such as Apple on 29 September 2000 (-0.73), the banks in 2008–09 and several stocks in March 2020, and none resembles an unadjusted split, so nothing was removed.

2 Returns are not Normal

Heavy tails. Excess kurtosis (zero for a Normal) is 114.6 for Apple and 89.0 for Procter & Gamble, driven by single days in 2000, and between 19 and 28 for BAC, WFC, HD, UNH and CVX; the S&P 500 has 10.6. The Jarque–Bera test rejects Normality for all 26 series. A Normal predicts 0.41 days beyond four standard deviations per series in the sample; the data have between 27 and 64, about 65 to 155 times as many. A Student-t fitted by maximum likelihood has between 2.2 and 3.6 degrees of freedom for every series (2.6 for the S&P 500), and and Figure 1 shows it follows the bulk and moderate tails of the index far better than a Normal, although the fit is imperfect in the far tails, where the few most extreme observations are smaller than the fitted t predicts. With four or fewer degrees of freedom the fourth moment is infinite, so sample kurtosis is not a stable quantity.

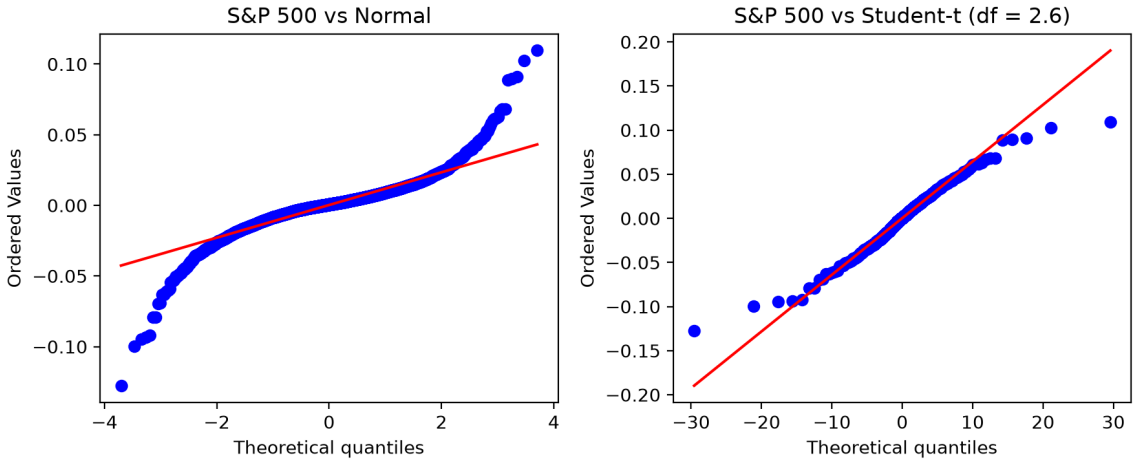


Figure 1: QQ plots of S&P 500 daily log returns against a Normal (left) and a fitted Student-t (right).

Volatility clustering. Returns themselves are almost uncorrelated: the average lag-1 autocorrelation is -0.049, which is statistically detectable in 25 of 26 series with 6,537 observations but explains about 0.25% of the variance of the next return. Squared returns are very different: the average lag-1 autocorrelation is 0.193 and the Ljung–Box test (10 lags) rejects independence for all 26 series. Large moves follow large moves; Figure 2 shows the slowly decaying autocorrelation of $|r|$ and the volatility regimes of 2000–02, 2008–09 and 2020.

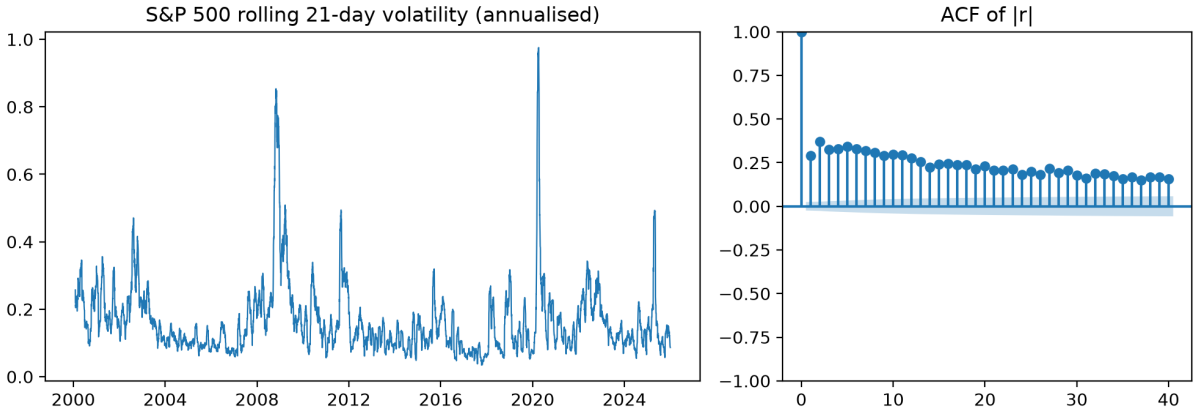


Figure 2: S&P 500 rolling 21-day volatility (left) and autocorrelation of absolute returns (right).

Correlations in crises. The average pairwise correlation of the 25 stocks is 0.26 in 2006–07 and 0.30 in 2019, but 0.63 from September 2008 to March 2009 and 0.72 from February to April 2020, while average volatility rises from about 20% to 72–80%. Part of this increase may come from selecting high-volatility windows rather than from a change in the underlying relationship [2], and the windows are short, so the figures are indicative.

3 Mean–variance optimisation

Let μ be the vector of expected returns, Σ the covariance matrix and w the weights. The minimum-variance portfolio with expected return m solves

$$\min_{w} \tfrac{1}{2} w^{\top} \Sigma w \quad \text{subject to} \quad \mathbf{1}^{\top} w = 1, \quad \mu^{\top} w = m.$$

The Lagrangian is $L = \frac{1}{2} w^\top \Sigma w - \lambda(\mathbf{1}^\top w - 1) - \gamma(\mu^\top w - m)$, and $\partial L/\partial w = \Sigma w - \lambda \mathbf{1} - \gamma \mu = 0$ gives $w = \Sigma^{-1}(\lambda \mathbf{1} + \gamma \mu)$. With $A = \mathbf{1}^\top \Sigma^{-1} \mathbf{1}$, $B = \mathbf{1}^\top \Sigma^{-1} \mu$, $C = \mu^\top \Sigma^{-1} \mu$ and $D = AC - B^2$, substituting into the two constraints gives $\lambda A + \gamma B = 1$ and $\lambda B + \gamma C = m$, so

$$\lambda = \frac{C - Bm}{D}, \qquad \gamma = \frac{Am - B}{D}.$$

Without the return constraint ($\gamma = 0$, $\lambda = 1/A$) the global minimum-variance portfolio is $w = \Sigma^{-1}\mathbf{1}/(\mathbf{1}^\top \Sigma^{-1} \mathbf{1})$. The variance along the frontier is $\sigma^2(m) = w^\top \Sigma w = \lambda + \gamma m = (Am^2 - 2Bm + C)/D$, a parabola in m, so the frontier is a hyperbola in (σ, m) with its leftmost point at $m = B/A$, $\sigma^2 = 1/A$. The maximum-Sharpe (tangency) portfolio satisfies $\Sigma w \propto (\mu - r_f \mathbf{1})$, hence $w \propto \Sigma^{-1}(\mu - r_f \mathbf{1})$ rescaled to sum to one. This is valid only when $\mathbf{1}^\top \Sigma^{-1}(\mu - r_f \mathbf{1}) > 0$, and the weights explode as that sum approaches zero.

The closed form was implemented with linear solves (never explicit inverses) and checked against a numerical solver (SLSQP): the largest difference in any weight on the 25 stocks was 5.1×10^{-6}. On the full sample, the minimum-variance portfolio has an annualised volatility of 14.9%, below every stock, with a return of 12.2%. The tangency portfolio (with $r_f = 0$) is extreme: 10 negative weights, weights from -18% to $+33\%$, and gross exposure of 3.05 (about 203% long and 103% short). Its weights follow the estimated means, whose standard error is about six percentage points for a stock with 30% volatility over 26 years, comparable to the whole spread of means in the sample.

4 Estimation error

Random matrix theory. For n observations of p independent variables with $q = p/n$, the eigenvalues of the sample correlation matrix fall between $(1 - \sqrt{q})^2$ and $(1 + \sqrt{q})^2$ (Marchenko–Pastur [4]). In the last 250 days ($q = 0.10$, upper edge 1.73) only two eigenvalues of the stock correlation matrix lie above the edge (7.94 and 3.41); in the full sample (upper edge 1.13) three do (9.31, 1.86 and 1.57). Everything inside the bulk is indistinguishable from noise (Figure 3).

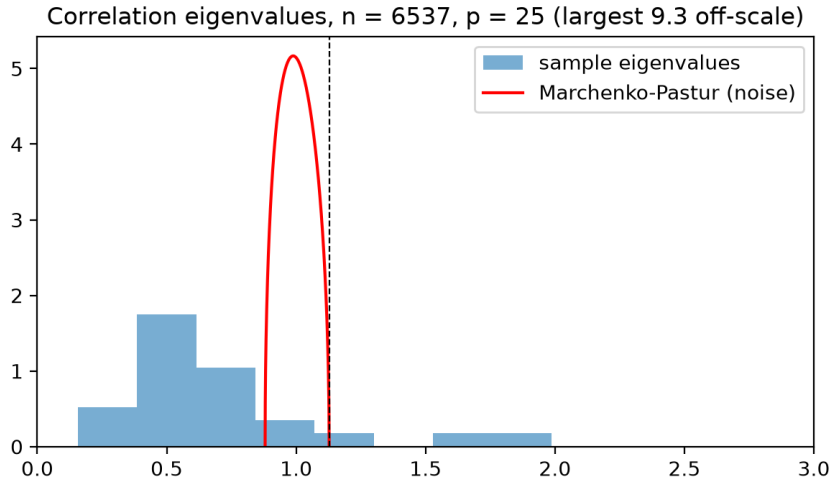


Figure 3: Eigenvalues of the sample correlation matrix (full sample) against the Marchenko–Pastur density for pure noise. The largest eigenvalue, the market factor, lies far to the right.

Shrinkage. The Ledoit–Wolf estimator [3] shrinks the sample covariance S towards a scaled identity, $\Sigma^* = \delta\, mI + (1 - \delta)S$ with $m = \mathrm{tr}(S)/p$ and a data-driven $\delta \in [0, 1]$. My implementation matches scikit-learn exactly. The intensity is $\delta = 0.139$ with 250 days and 0.006 with the full sample, and the condition number falls from 109 to 34 in the short window.

Simulation with a known truth. I drew data from a one-factor covariance matrix with 25 assets (500 repetitions per n) and, for each estimate, built minimum-variance weights and measured their true variance relative to the best achievable. Table 1 shows that estimation error costs 69% extra variance at $n = 60$ with the sample covariance, 34% with Ledoit–Wolf, and falls towards zero as n grows. The EWMA estimate ($\lambda = 0.94$, effective sample about 17 days) stays near 1.95 at every n because its effective sample is smaller than the number of assets and the simulated covariance is constant.

Table 1: True variance of minimum-variance portfolios relative to the optimum (1 is perfect).

n	Sample	Ledoit–Wolf	EWMA
60	1.689	1.335	1.984
120	1.259	1.196	1.948
250	1.109	1.098	1.941
500	1.051	1.048	1.954
1000	1.024	1.023	1.939

5 Out-of-sample backtest

Design. Each strategy is run walk-forward on the simple returns of the 25 stocks. Every 21 trading days the weights are computed from the previous 504 days only (rows $t-504, \ldots, t-1$ for a rebalance on day t) and applied from day t; between rebalances they drift with returns. A transaction cost of 10 basis points of turnover is deducted. The settings were fixed before looking at results. The strategies are 1/N; minimum variance with the sample covariance; minimum variance with the Ledoit–Wolf covariance, with and without a long-only constraint; and a tangency-type portfolio using Ledoit–Wolf covariance and means shrunk halfway to their cross-sectional average ($k = 0.5$). The last falls back to minimum variance when the tangency formula is invalid or gross exposure would exceed 3, which happened at 152 of 288 rebalances. Tests check that changing a future return cannot alter any earlier result.

Table 2: Backtest results, net of 10 bps costs (window 504 days, monthly rebalancing).

Strategy	Return	Vol.	Sharpe	Max DD	Turnover/yr	Gross Sharpe
1/N	14.6%	18.6%	0.823	−43.8%	0.58	0.826
MinVar, sample	7.5%	14.5%	0.571	−40.2%	3.30	0.594
MinVar, LW	8.0%	14.4%	0.609	−37.5%	2.83	0.629
MinVar, LW long-only	9.5%	14.5%	0.698	−35.3%	1.47	0.708
Shrunk MV	11.0%	16.3%	0.725	−37.5%	8.43	0.777

Results. Equal weighting has the best Sharpe ratio (0.823) and ranks first in every sensitivity setting (Table 3), although it also has the deepest drawdown. The minimum-variance strategies lower volatility from 18.6% to about 14.5% but give up more return than they save in risk. Ledoit–Wolf beats the sample covariance in all four settings (by 0.03 to 0.06 of Sharpe) with lower turnover, and the long-only constraint helps more still (0.698 against 0.609, half the turnover). The shrunk mean–variance strategy has the highest turnover by far (8.4 a year), loses 0.05 of Sharpe to costs, and its rank moves between second and fourth. In sub-periods, 1/N's lead comes largely from 2020–2025 (Table 4); in 2010–2019 all five strategies lie between 1.10 and 1.20.

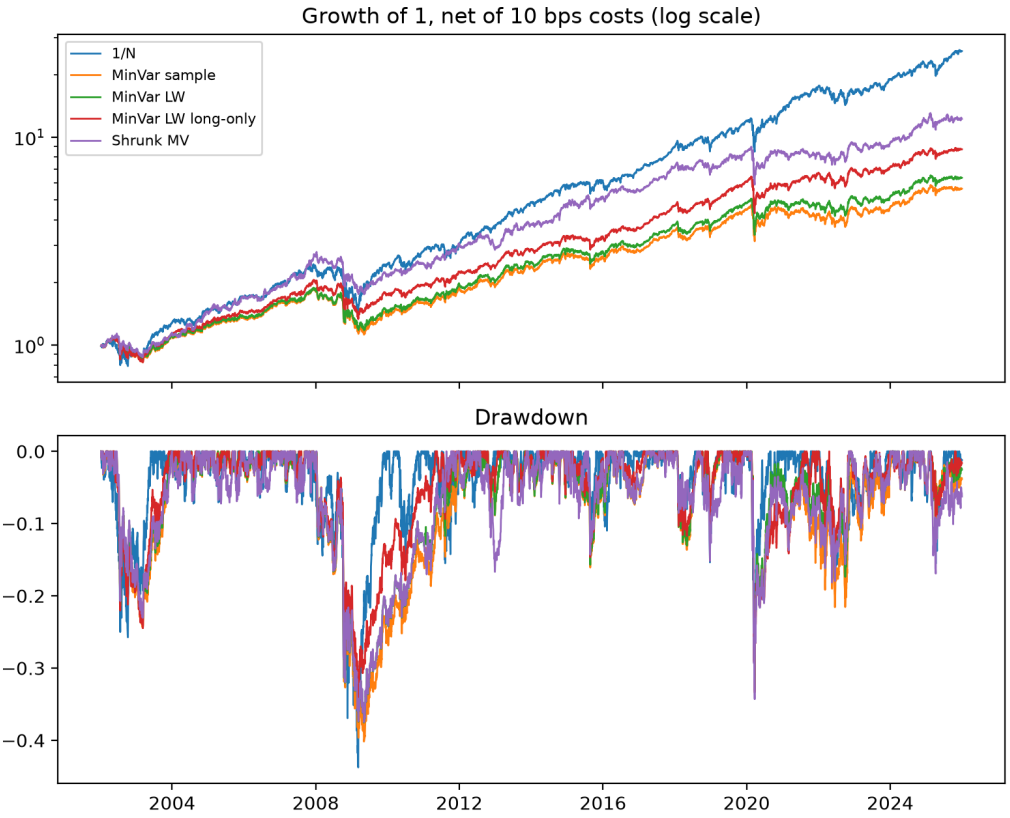


Figure 4: Walk-forward wealth and drawdown of the five strategies, net of costs.

Table 3: Net Sharpe ratio under alternative settings (window / rebalancing days / cost in bps).

Setting	1/N	MinVar sample	MinVar LW	LW long-only	Shrunk MV
252 / 21 / 10	0.805	0.566	0.625	0.698	0.621
504 / 63 / 10	0.827	0.579	0.613	0.699	0.745
504 / 21 / 0	0.826	0.594	0.629	0.708	0.777
504 / 21 / 30	0.817	0.526	0.570	0.677	0.620

How much weight these differences carry. Over 23.9 years the approximate standard error of each Sharpe ratio is 0.22 to 0.24 (an i.i.d. approximation). The lead of 1/N over the other strategies is 0.10 to 0.25, so at most about one standard error. The strategies are correlated, which makes differences more precise than this, but I did not test that, so the data show that nothing beat 1/N, not that 1/N is better.

6 Conclusion and limitations

Real returns have heavy tails, clustered volatility and crisis correlations, and these features make sample means and covariances unreliable inputs. Shrinkage and constraints reduce the damage, in line with the evidence that estimation error limits the gains from optimisation [1], but a simple equal-weight rule was not beaten here.

The limitations are the following. (i) *Survivorship bias*: the stocks are today's large companies, so failures are missing and results are flattered, possibly 1/N more than the others. (ii) One market, one universe of 25 stocks and one 26-year period. (iii) The simulation assumes Normal data and a constant one-factor covariance. (iv) Sharpe ratios have large standard errors. (v) Window, rebalancing, cost and shrinkage constants were fixed in advance, and the sensitivity

Table 4: Net Sharpe ratio by sub-period (main settings).

Period	1/N	MinVar sample	MinVar LW	LW long-only	Shrunk MV
2002–2009	0.606	0.337	0.370	0.509	0.643
2010–2019	1.188	1.122	1.157	1.204	1.100
2020–2025	0.768	0.308	0.357	0.438	0.411

tables show how much rankings move. (vi) Costs are proportional to turnover only: no market impact, borrowing cost for shorts or taxes, and the risk-free rate is zero. (vii) The shrunk mean–variance strategy is partly minimum variance.

References

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